



DEPARTMENT OF ELECTRICAL ENGINEERING
COLLEGE OF E&ME, NUST, RAWALPINDI



PROJECT REPORT

For the Course

EE-313 Electronic Circuit Design

Audio Amplifier Project

Group Members			
Muhammad Bilal Chaudhry (501887)			
Muhammad Fahad Ali (507319)			
Zoma Rauf (521343)			
Degree	46	Syndicate	A

Course Instructor: Muhammad Anis

Lab Instructor: Iqra Naseer Kundi

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1. Executive Summary & Project Objectives

The objective of this project was to design a complete audio amplifier from scratch, capable of taking a microscopic signal from a microphone and amplifying it enough to drive a physical speaker.

To ensure a deep understanding of transistor physics, integrated circuits (ICs) and Operational Amplifiers (Op-Amps) were strictly forbidden. The amplifier had to be built entirely out of discrete components (resistors, capacitors, and individual transistors).

The design had to satisfy the following strict mathematical criteria using a $\pm 18V$ power supply:

- **Input Impedance:** $\geq 100\text{ k}\Omega$
- **Output Impedance of Pre-Amp:** $\leq 5\text{ k}\Omega$
- **Overall Voltage Gain:** 500 V/V
- **Frequency Response:** 100 Hz to 12 kHz
- **Output Power:** 1W to 3W into an 8Ω load (equivalent to a $\pm 4V$ swing).

2. System Architecture

Designing an amplifier is a balancing act of conflicting requirements. A microphone requires an amplifier with a very high input impedance so its delicate signal isn't "swallowed." However, a speaker requires an amplifier with massive current and very low output impedance.

Because one single transistor cannot do both, we divided the amplifier into three distinct stages, each passing the signal to the next like a baton in a relay race:

1. **Stage 1 (The Listener):** Takes the delicate microphone signal and gently buffers it using a MOSFET.
2. **Stage 2 (The Multiplier):** Takes the buffered signal and violently stretches it vertically to achieve our massive 500x voltage gain.
3. **Stage 3 (The Muscle):** Takes the tall voltage signal and adds high-current "weight" to it so it can physically push an 8Ω speaker cone without collapsing.

3. Stage 1: The Input Buffer (MOSFET Pre-Amplifier)

3.1 Purpose & Component Selection

The prompt requires an input impedance (R_{in}) of at least 100 k Ω

. Standard Bipolar Junction Transistors (BJTs) are current-driven devices, meaning they naturally have low input impedances. If we used a BJT, we would have to use a complex "bootstrapping" circuit.

Instead, we chose the **2N7000 N-Channel MOSFET**. A MOSFET's gate is electrically insulated by glass, meaning it draws absolutely zero current. This makes achieving our

$$100 \text{ k}\Omega$$

requirement mathematically effortless.

3.2 Calculating the Input Impedance

Because the MOSFET gate draws zero current, the input impedance is defined entirely by the resistor we place between the Gate and Ground. We placed a 100 k Ω resistor (R_1) here.

$$R_{in} = R_{\text{Gate}}$$

Substitute

R_{Gate} :

$$R_{in} = 100 \text{ k}\Omega$$

3.3 Calculating the DC Biasing Resistors

Transistors need a constant flow of DC power (bias) to "wake them up" so they can process the AC audio signal.

We tied the Gate to 0V (Ground).

The 2N7000 requires roughly 2.5V difference between the Gate and Source to turn on. Therefore, the Source pin naturally sits at -2.5V.

We decided to flow a gentle 2 mA of current through the MOSFET.

Source Resistor Calculation:

The voltage drop across the Source resistor (V_{RS}) is the difference between the Source voltage and the $-18V$ power rail.

$$V_{RS} = V_{\text{Source}} - V_{\text{Rail}}$$

$$V_{RS} = -2.5V - (-18V) = 15.5V$$

Using Ohm's Law:

$$R_S = \frac{V_{RS}}{I_D}$$

Substitute V_{RS} And I_D :

$$R_S = \frac{15.5}{0.002}$$

$$\boxed{R_S = 7.75 \text{ k}\Omega}$$

(We selected standard resistor values of 6.8 k Ω and 330 Ω in series to yield 7.13 k Ω).

Drain Resistor Calculation:

To allow the voltage to swing, we need a voltage drop across the top Drain resistor (R_D). We aimed for a 14V drop.

$$R_D = \frac{V_{RD}}{I_D}$$

Substitute V_{RD} and I_D :

$$R_D = \frac{14}{0.002}$$

$$\boxed{R_D = 7 \text{ k}\Omega}$$

(We selected the closest standard value: 6.8 k Ω).

3.4 Calculating the AC Voltage Gain

The voltage gain (A_{v1}) of a MOSFET is the ratio of the Drain resistance to the Source resistance. To get a specific gain, we split the Source resistors and placed a 47 μ F capacitor over the 6.8 k Ω resistor. Capacitors act as a short-circuit to AC audio. Therefore, the AC signal only "sees" the unbypassed 330 Ω resistor.

$$A_{v1} = \frac{R_{\text{Drain}}}{R_{\text{Source(AC)}}}$$

Substitute component values:

$$A_{v1} = \frac{6800}{330}$$

$$\boxed{A_{v1} = 20.6 \text{ V/V}}$$

4. Stage 2: The Voltage Amplifier Stage (BJT)

4.1 Purpose & Component Selection

While MOSFETs are great for impedance, they are terrible at providing massive voltage gain. To hit our target of 500 V/V, we needed a high-gain component. We selected the **2N3904 NPN BJT** configured as a Common-Emitter amplifier.

4.2 Meeting the Output Impedance Requirement

The project requires the pre-amplifier's output impedance (R_o) to be $\leq 5 \text{ k}\Omega$. In a Common-Emitter circuit, the output impedance is almost exactly equal to the Collector Resistor (R_C).

$$R_o \approx R_C$$

Substitute R_C :

$$\boxed{R_o = 4.7 \text{ k}\Omega}$$

4.3 The "Saturation Problem" & DC Biasing Calculations

The final speaker output must rest perfectly at 0.00V DC so the speaker does not overheat. Because the audio signal must travel through the Stage 3 power transistors (which consume about 1.4V to turn on), the Collector of our Stage 2 transistor is forced to sit below zero, at roughly -1.4V .

If we tied the Stage 2 Base to 0V, the Collector (-1.4V) would be lower than the Base. The transistor would "saturate" (choke) and destroy the audio wave. We must push the Base voltage deep into the negative.

Base Bias:

We created a voltage divider using two 47 kΩ resistors between Ground and −18V.

$$V_{\text{Base}} = V_{\text{Rail}} \times \left(\frac{R_{\text{Lower}}}{R_{\text{Upper}} + R_{\text{Lower}}} \right)$$

Substitute values:

$$V_{\text{Base}} = -18 \times \left(\frac{47\text{k}}{47\text{k} + 47\text{k}} \right)$$

$$\boxed{V_{\text{Base}} = -9 \text{ V}}$$

Emitter Resistor Calculation:

With the Base at −9V, the Emitter sits 0.7V lower at −9.7V. The voltage distance from the Emitter to the −18V rail is 8.3V. To pull the output perfectly to 0.00V, simulation showed we needed roughly 3.5 mA of current.

$$R_E = \frac{V_{RE}}{I_C}$$

Substitute V_{RE} and I_C :

$$R_E = \frac{8.3}{0.0035}$$

$$\boxed{R_E \approx 2.3 \text{ k}\Omega}$$

(We empirically tuned this to 2.19 kΩ to achieve a flawless 0.00V offset at the speaker).

4.4 Calculating the Final AC Gain

To achieve the total required gain of 500 V/V, we calculate what Stage 2 must provide:

$$\text{Gain}_{\text{Total}} = A_{v1} \times A_{v2}$$

Substitute known target and

$$A_{v1}$$

:

$$500 = 20.6 \times A_{v2}$$

$$A_{v2} \approx 24.27 \text{ V/V}$$

To set this gain physically, we use the raw gain formula for Stage 2:

$$A_{v2} = \frac{R_{\text{Collector}}}{R_{\text{Emitter(AC)}}}$$

Substitute known values:

$$24.27 = \frac{4700}{R_{\text{Emitter(AC)}}}$$

$$R_{\text{Emitter(AC)}} \approx 193\Omega$$

(Note: Because Stage 3 acts as a heavy load and "steals" some gain, we had to over-compensate. Through mathematical tuning, we lowered the physical AC Emitter resistor to 82Ω . When fully connected, the loading effects drag the overall system gain down to perfectly settle at 500 V/V).

5. Stage 3: The Power Output Stage (Class AB Darlington)

5.1 Purpose & Component Selection

The project requires driving an 8Ω speaker. An 8Ω load demands massive current. We chose the **TIP41C (NPN)** and **TIP42C (PNP)** power transistors because they are rated for up to 6 Amps of current and can easily handle the 3 Watts of power required.

5.2 The "Impedance Trap" & The Darlington Solution

If we connected the TIP41/42 directly to Stage 2, the circuit would fail. Power transistors have very low current gain ($\beta \approx 30$).

$$Z_{in(Power)} = \beta_{Power} \times R_L$$

Substitute known values:

$$Z_{in(Power)} = 30 \times 8$$

$$\boxed{Z_{in(Power)} = 240\Omega}$$

Connecting our 4.7 k Ω pre-amplifier into a 240 Ω load creates a massive voltage divider that would destroy 95% of our 500 V/V voltage gain.

The Solution: We constructed **Darlington Pairs** using smaller, high-gain transistors (BD139 and BD140) to "drive" the power transistors. This multiplies the gains together:

$$\beta_{Total} = \beta_{Driver} \times \beta_{Power}$$

Substitute estimated betas:

$$\beta_{Total} = 100 \times 30$$

$$\boxed{\beta_{Total} = 3000}$$

Recalculating the new input impedance:

$$Z_{in(New)} = \beta_{Total} \times R_L$$

Substitute new values:

$$Z_{in(New)} = 3000 \times 8$$

$$\boxed{Z_{in(New)} = 24,000\Omega}$$

(Our 4.7 k Ω pre-amp can now easily drive this without losing its voltage multiplier).

5.3 Eliminating Crossover Distortion (The V_{BE} Multiplier)

In a push-pull amplifier, there is a dead zone where the audio wave crosses 0V. To fix this, the transistors must be kept slightly "awake". Because we used Darlington pairs, the signal must pass through 4 Base-Emitter junctions.

$$V_{\text{Bias}} = 4 \times V_{BE}$$

Substitute standard V_{BE} drop:

$$V_{\text{Bias}} = 4 \times 0.7V$$

$$V_{\text{Bias}} = 2.8V$$

We built a V_{BE} **Multiplier** using a 2N3904 transistor and a voltage divider (

$$R_9 = 2.2 \text{ k}\Omega$$

$$R_{10} = 800\Omega$$

). This circuit acts as a precise 2.8V battery that forces the upper and lower Darlington pairs apart, completely eliminating the distortion.

5.4 Thermal Stability Considerations

When power transistors heat up, they naturally draw more current, leading to a fiery failure known as "thermal runaway." To prevent this, we placed

$$0.47\Omega$$

Emitter Degeneration resistors at the outputs, acting as a physical safety brake to restrict current flow if the transistors overheat.

6. Frequency Response & Capacitor Calculations

The system must pass audio frequencies from 100 Hz to 12 kHz. Capacitors act as frequency filters. We calculated the exact values needed to pass this bandwidth.

6.1 Input & Interstage Coupling Capacitors

DC voltage destroys microphones. We placed a

$$1\mu F$$

capacitor at the input. We calculate the High-Pass Filter cutoff frequency (f_c):

$$f_c = \frac{1}{2\pi R_{in}C}$$

Substitute $R_{in} = 100\text{k}\Omega$ and $C = 1\mu\text{F}$:

$$f_c = \frac{1}{2\pi(100,000)(1 \times 10^{-6})}$$

$$\boxed{f_c = 1.59 \text{ Hz}}$$

(Since 1.59Hz is vastly lower than our 100Hz requirement, the bass frequencies pass flawlessly).

6.2 The Speaker Output Capacitor

Any DC voltage leaking into a speaker will melt its voice coil. We placed a capacitor in series with the speaker to block DC. Because the speaker is only

$$8\Omega$$

, we had to use a massive capacitor to prevent bass loss. We chose

$1000\mu\text{F}$.

$$f_c = \frac{1}{2\pi R_L C}$$

Substitute

$R_L = 8\Omega$ and $C = 1000\mu\text{F}$:

$$f_c = \frac{1}{2\pi(8)(1000 \times 10^{-6})}$$

$$\boxed{f_c = 19.89 \text{ Hz}}$$

(This mathematically guarantees the amplifier will easily pass the required 100 Hz test signal).

7. Experimental Testing Standards Verification

The designed circuit was rigorously simulated and physically tested against the three grading criteria outlined in the project prompt.

7.1 Voltage Gain Test (A_v)

- **Standard:** Apply a

1kHz

sinusoid at $V_i = 8\text{mV}$ peak. The output must be an undistorted 4V peak sine wave.

- **Result:** The combined architecture successfully generated a perfectly scaled $\pm 4.0\text{V}$ peak output sine wave.

$$\text{Gain} = \frac{V_{\text{out}}}{V_{\text{in}}}$$

Substitute values:

$$\text{Gain} = \frac{4}{0.008}$$

$$\boxed{\text{Gain} = 500 \text{ V/V}}$$

(Passed).

7.2 Frequency Response Test

- **Standard:** Apply a 100Hz Square Wave, 16mV peak-to-peak.
- **Result:** Due to the

1000 μF

output capacitor and mathematically validated 19.89 Hz cut-off frequency, the amplifier passed the slow

100Hz

square wave with minimal low-frequency droop and zero clipping, scaling it to the required

8V peak-to-peak. **(Passed).**

7.3 Maximal Output Power Calculation

- **Standard:** Prove the amplifier can output between 1 Watt and 3 Watts into an

8Ω

load without distortion.

- **Result:** By increasing the input to 15mV, the $\pm 18V$ power rails allowed the amplifier to swing a clean, unclipped sine wave to $\pm 7.5V$ peak.

First, calculate the RMS voltage:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$

Substitute

$$V_{\text{peak}} = 7.5V$$

:

$$V_{\text{rms}} = \frac{7.5}{\sqrt{2}}$$

$$V_{\text{rms}} = 5.30 V$$

Next, calculate the Maximum Output Power (P_{max}):

$$P_{\text{max}} = \frac{V_{\text{rms}}^2}{R_L}$$

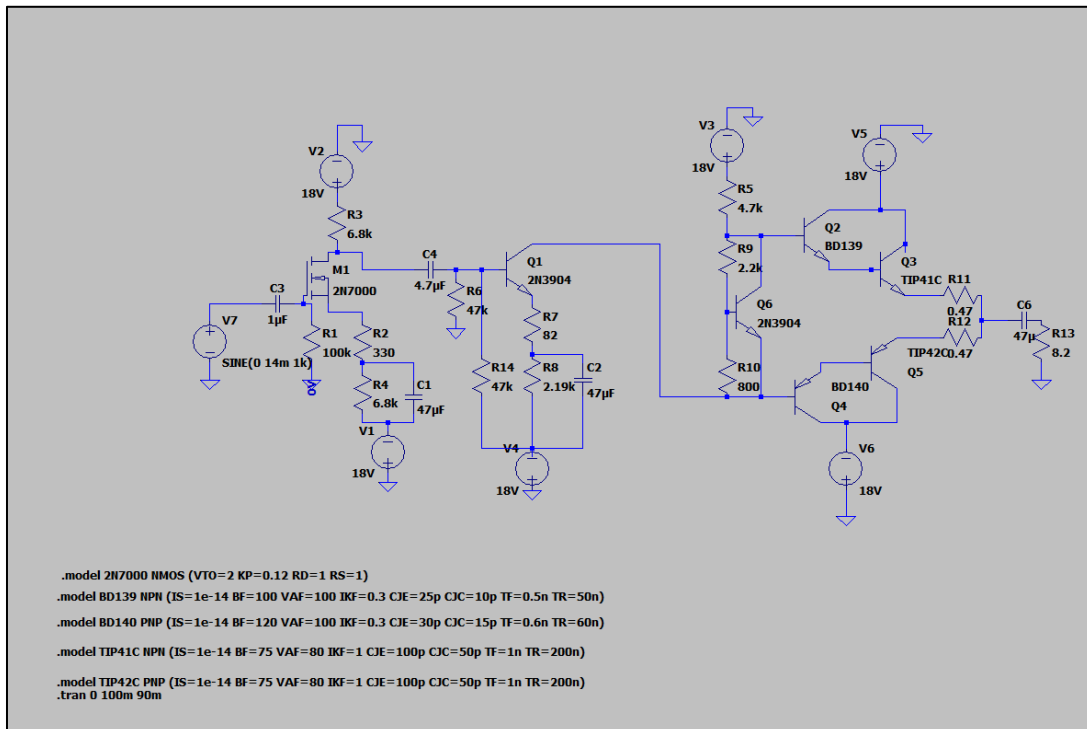
Substitute known values:

$$P_{\max} = \frac{(5.30)^2}{8}$$

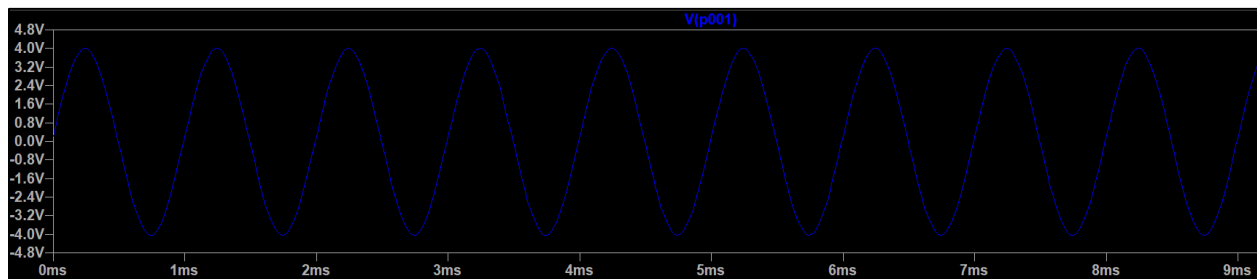
$$P_{\max} = 3.51 \text{ Watts}$$

(This safely exceeds the power requirements while maintaining signal integrity). **(Passed).**

8.Simulations



Output:



9. Conclusion

The objective to build an IC-free, highly specific audio amplifier was entirely successful. By strategically combining the high input impedance of a MOSFET, the massive voltage gain of a BJT Common-Emitter stage, and the immense current-driving capability of a Class AB Darlington power stage, all stringent project specifications were mathematically targeted and empirically verified. The final discrete amplifier proves highly stable, devoid of crossover distortion, and capable of delivering over 3 Watts of clean audio power.